

Product Design and Innovation

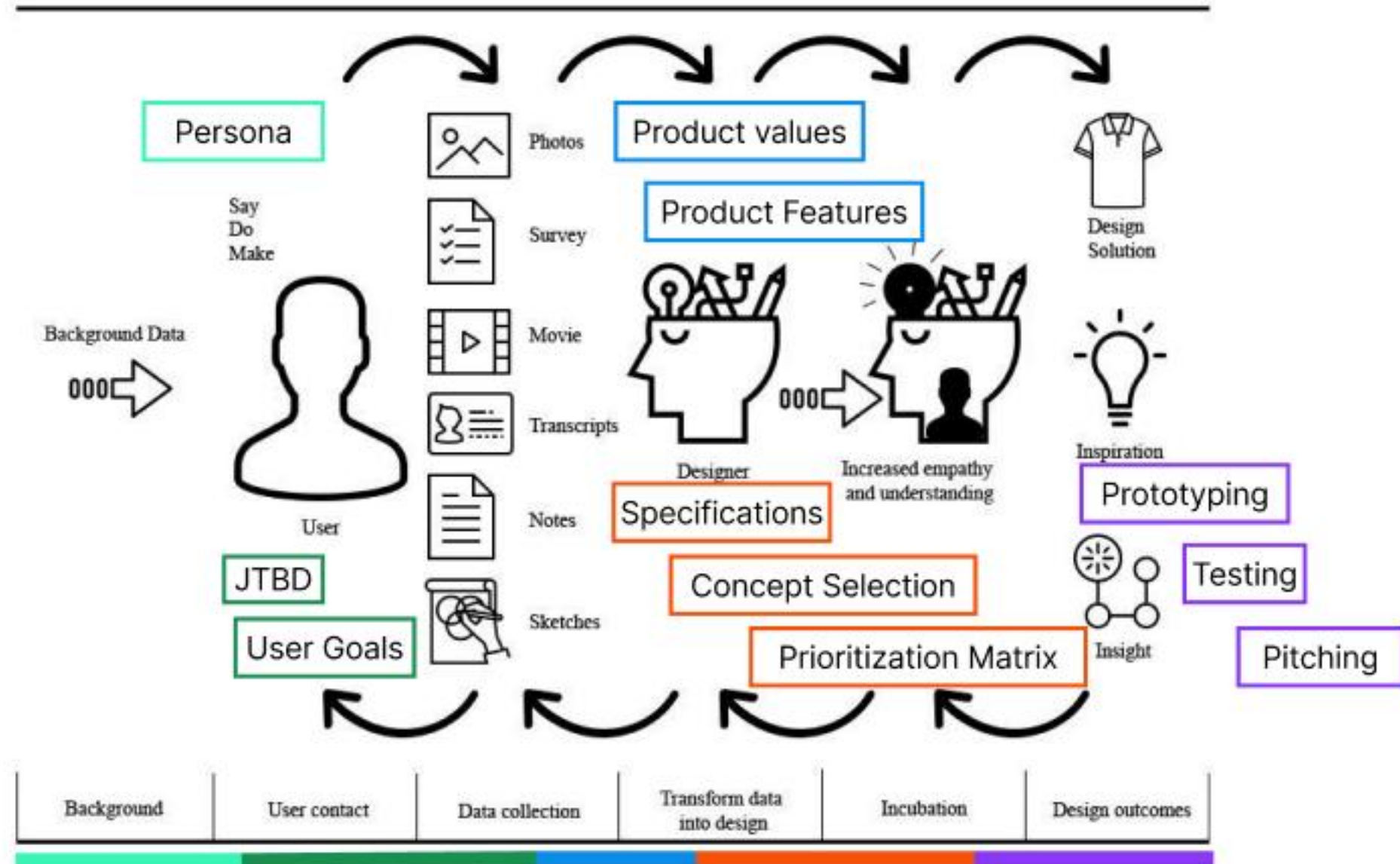


CDE2300 pdi



@CDE2300 Product Design and Innovation

Overview of Design Process

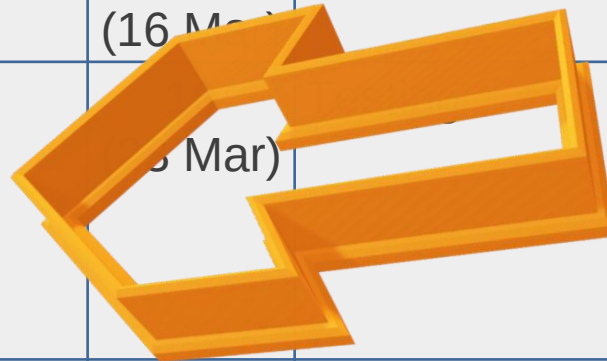


Course Overview

Week	Titles	Week	Titles
1 (12 Jan)	Innovation-driven economy, Group formation, and Brief intro	7 (2 Mar)	Concept Generation, Prioritization Map, Concept Selection, and Specifications
2 (19 Jan)	Experience economy, JTBD, Job Story and Job Statement	8 (9 Mar)	Prototyping and Prototyping Plan
3 (26 Jan)	Empathy Map, Fieldwork, and Persona	9 (16 Mar)	Storytelling
4 (2 Feb)	Use of AI in Design, Affinity Clustering, Ideation Methods, Users' Goals, Product Values and Features, and Hands-on Exercise: Crazy 8	10 (23 Mar)	Testing
5 (9 Feb)	Mid Term Review	11 (30 Mar)	Group consultation
6 (16 Feb)	No class!	12 (6 Apr)	Group consultation
23 Feb to 27 Feb	Recess Week (No Class)	13 (13 Apr)	End Term Review (Show and Tell) (2pm to 6pm)

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Assessment

Individual

- **Peer Review + Assessment (TAs) (5%)**
 - 2.5% (Mid Term), 2.5% (End Term)
 - Communication
 - Collaboration
 - Responsibility & Reliability
 - Proactive Problem Solving
 - Constructive Feedback
- **Design Reflection (due on Reading Week) (5%)**
- **Assignments (3 Homework Assignments) (5%)**

Group

- **End Term Deliverables (73%)**
- **Mid Term Deliverables (12%)**

Assessment

Mid Term Deliverables (12%)

- JTBD
- Summary of Fieldwork
- Affinity Clusters
- Persona(s)
- Product Values and Product Features

Each team is given 6 minutes to pitch and 4 minutes for Q&A. Random order. Rubrics out on 30 Jan (Friday).

To Prepare for your Fieldwork

Use the 5W and 1H method.

Who

are your
target
users?

What

do you
want to
find out?

Why

is it
important?
*(About the
Question)*

When

are you going
to conduct
the fieldwork

Where

will the
fieldwork
take place?

How

are you going to
reach out to them?
How are you going
to conduct the
fieldwork?

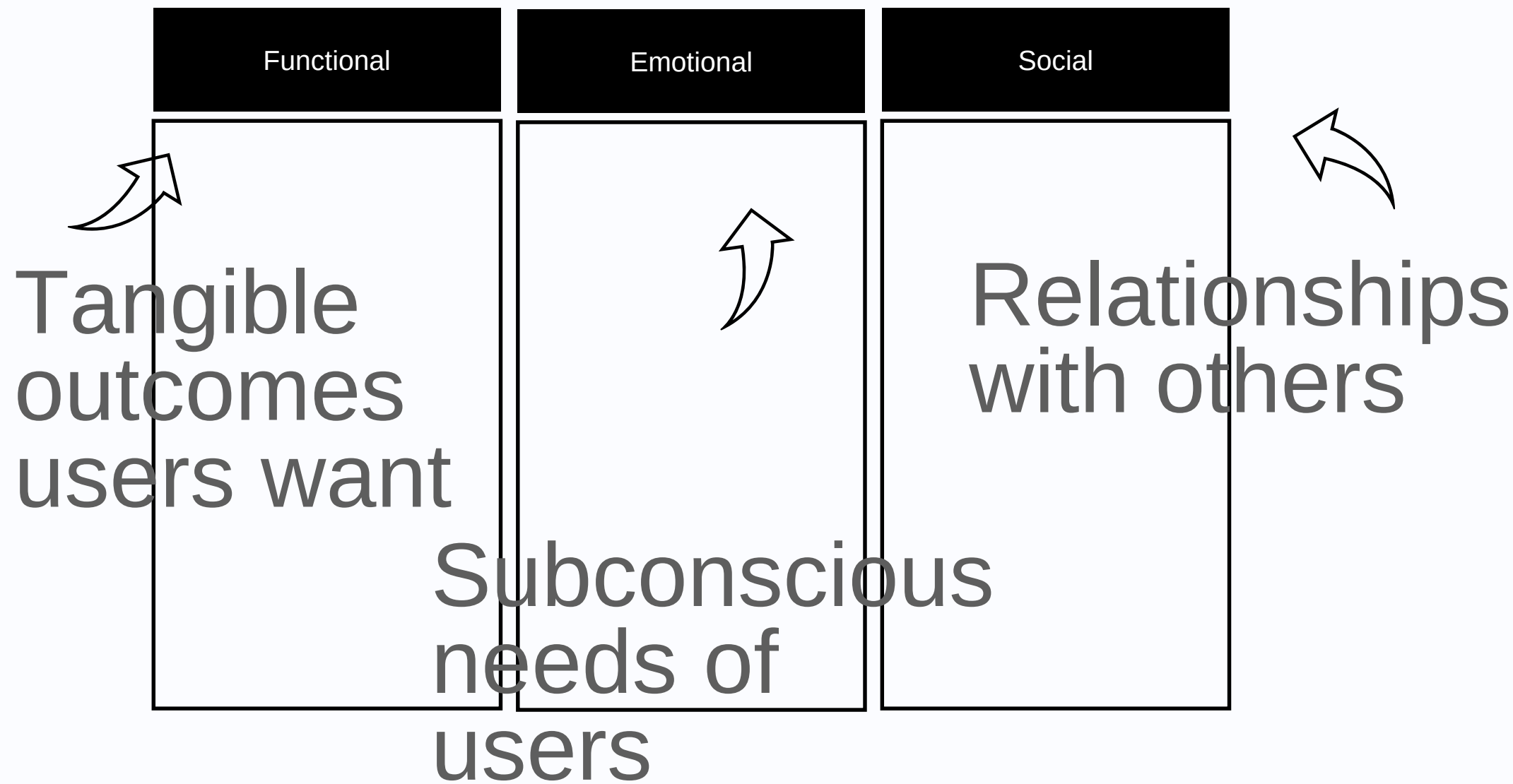
BEFORE
the Fieldwork

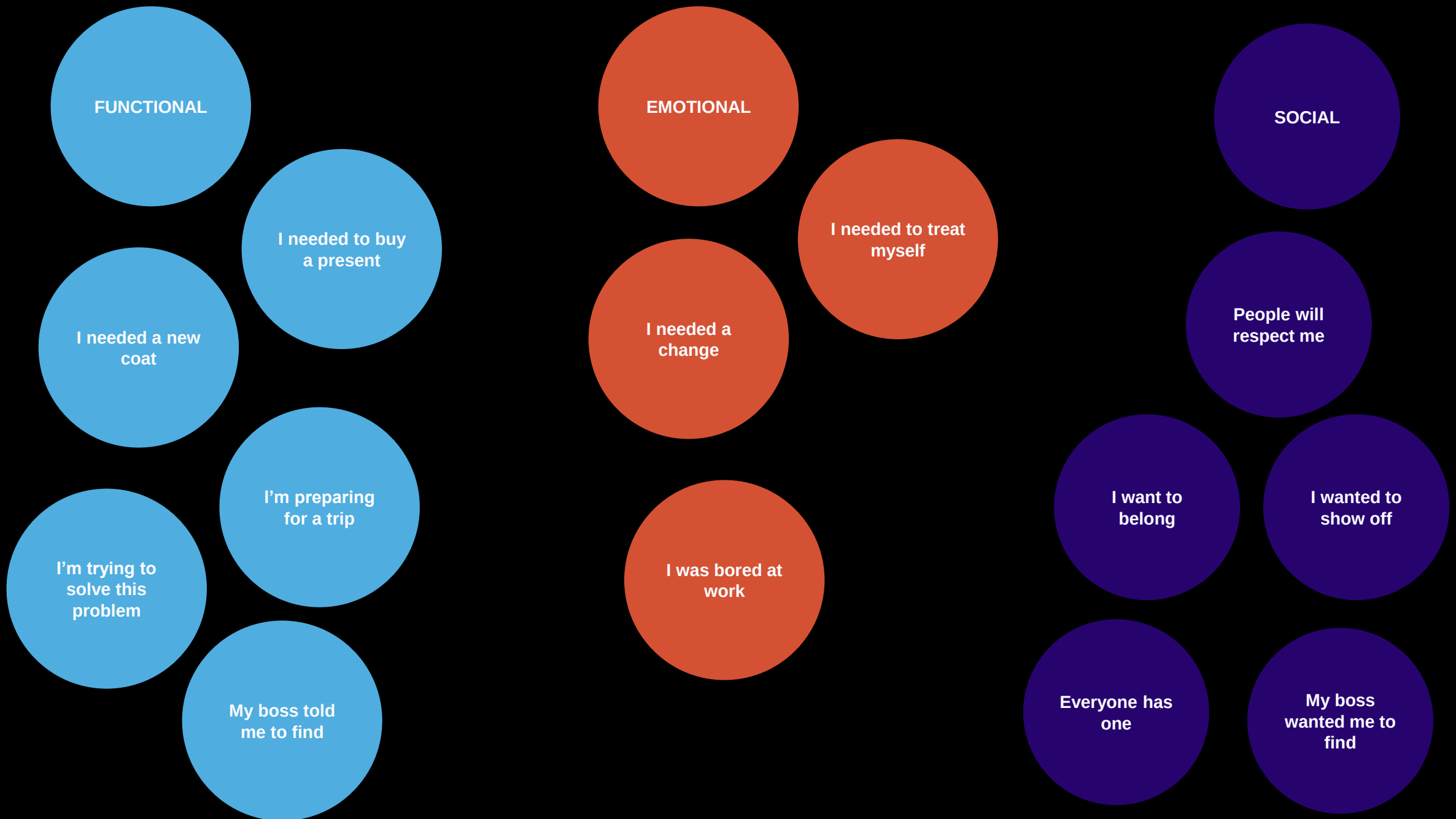


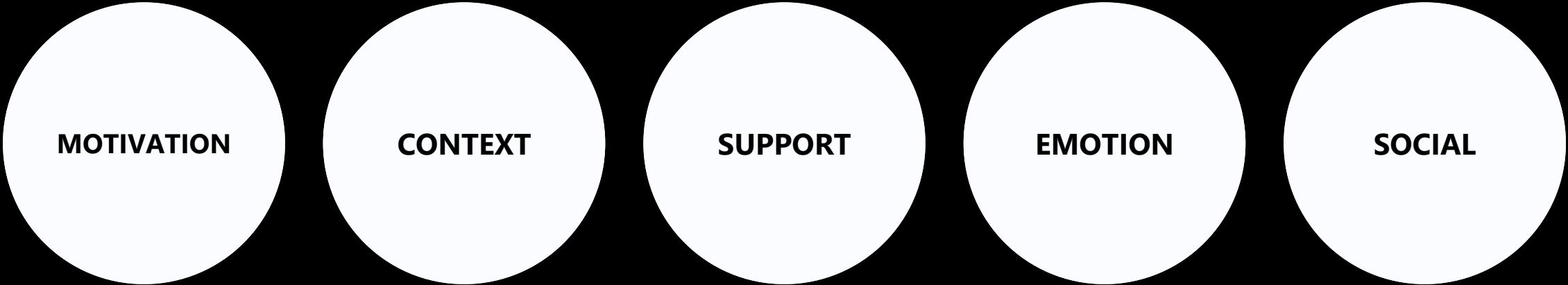
Understand customer needs (value)



JTBD Template







MOTIVATION

CONTEXT

SUPPORT

EMOTION

SOCIAL

Job Statement

WHEN I BUT HELP ME SO I



What's the context?



What's the barrier
getting in the way
today?



What is the user
trying to do?



What's the outcome of
achieving this action?

INTERVIEW your partner about a purchase

a **significant purchase** they had made in the last few months

“What have you bought recently?”

- *INTERVIEW by using JTBD framework. 20 min*
- *Write a **Job Statement**. 5 min*



CHOOSE A
PURCHASE
THAT
MATTERED IN
THE LAST 3
MONTHS

MOTIVATION

**When did you
first think
about buying
something
like this?**

Interviewee:

**Describe when you first thought about
buying something in this category.**

Interviewer:

Listen for the needs.

CONTEXT



Tell me the story about buying this product.

Interviewee:

Describe the process of buying the product in rich detail. Help us understand each step.

Interviewer:

Listen to the story. Spot emotional decisions made.

SUPPORT

**Why did you
buy this
product?**

Interviewee:

**Describe why you bought this product.
What it is for. What it does for you.**

Interviewer:

Capture obvious jobs.

EMOTION



How do you
feel about
your purchase
now?

Interviewee:

Describe how this product or service makes you feel. How well it solves your needs.

Interviewer:

Listen for future opportunities.

SOCIAL

How did you
share your
buying story
with friends
and family?

Interviewee:

Did you tell other people about your purchase? How did they react? How did that make you feel?

Interviewer:

Listen for social jobs.

Job Statement (Example)



WHEN I want to recruit new participants for CDE2300 project, **BUT I** have so many assignments to completed, **HELP ME** to automate the process of recruitment, screening, scheduling and reminder, **SO I** can quickly collect lots of data from useful participants without the enormous time spent on doing it manually.



Nerdy Nina

"The book is way better than the movie!"

#booklover

#bookaddict

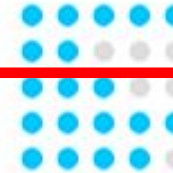
#bookloverproblems

DEMOGRAPHICS

Age: 25
Location: Sao Paulo, Brazil
Education: Software Engineer
Job: Q/A at Indie Game Company
Family: Lives with her boyfriend

TECH

Internet
Social Networks
Messaging
Games
Online Shopping



GOALS

- Discovering new books / authors to read
- Finding unique stories
- Cataloging book collection

FRUSTRATIONS

- Keeping track of different series
- Forgetting a book launch date
- Finding space for more books

READING HABITS

- Fast pace reader
- Never lends books
- Likes hardcovers and boxed collections
- Pre-order books to get them first
- Reads eBooks, but prefer physical copies
- Always finishes a book
- Loves binge reading and re-reading

FAVORITE BOOKS



American Gods
Neil Gaiman



Harry Potter
J.K. Rowling



Ready Player
One

Empathic Title

Quote

Basic Demographics

Goals & Frustrations

Habits/Preferences

A man with a shaved head, wearing a grey and white striped long-sleeved shirt, stands in a kitchen. The kitchen has wooden upper cabinets and a white tiled backsplash. On the counter in front of him are various condiments, including several bottles of sauce and a large red carton of '1 liter' of something. A dashed white circle highlights a group of small sauce bottles on the left. Above the man, on a shelf, are various food items, including a large orange container and several boxes. The man is looking slightly to his left.

Odi, 26
Surabaya

Odi creates
his own sauce

"When I make fried noodles I use tomato sauce and other sauces to make my own creation. I also use Saos Raja Rasa for fried rice."



Easy-going-about-food Iqbal

"We cook rice. We buy all other dishes. No time to cook."

Demographic

Age: 35 years old

Occupation: employee of a company

Family: married with 2 kids

Spouse: working

Experience with food

- Seldom cook at home
- Lunch: eat out near workplace
- Dinner: cook rice and buy dishes

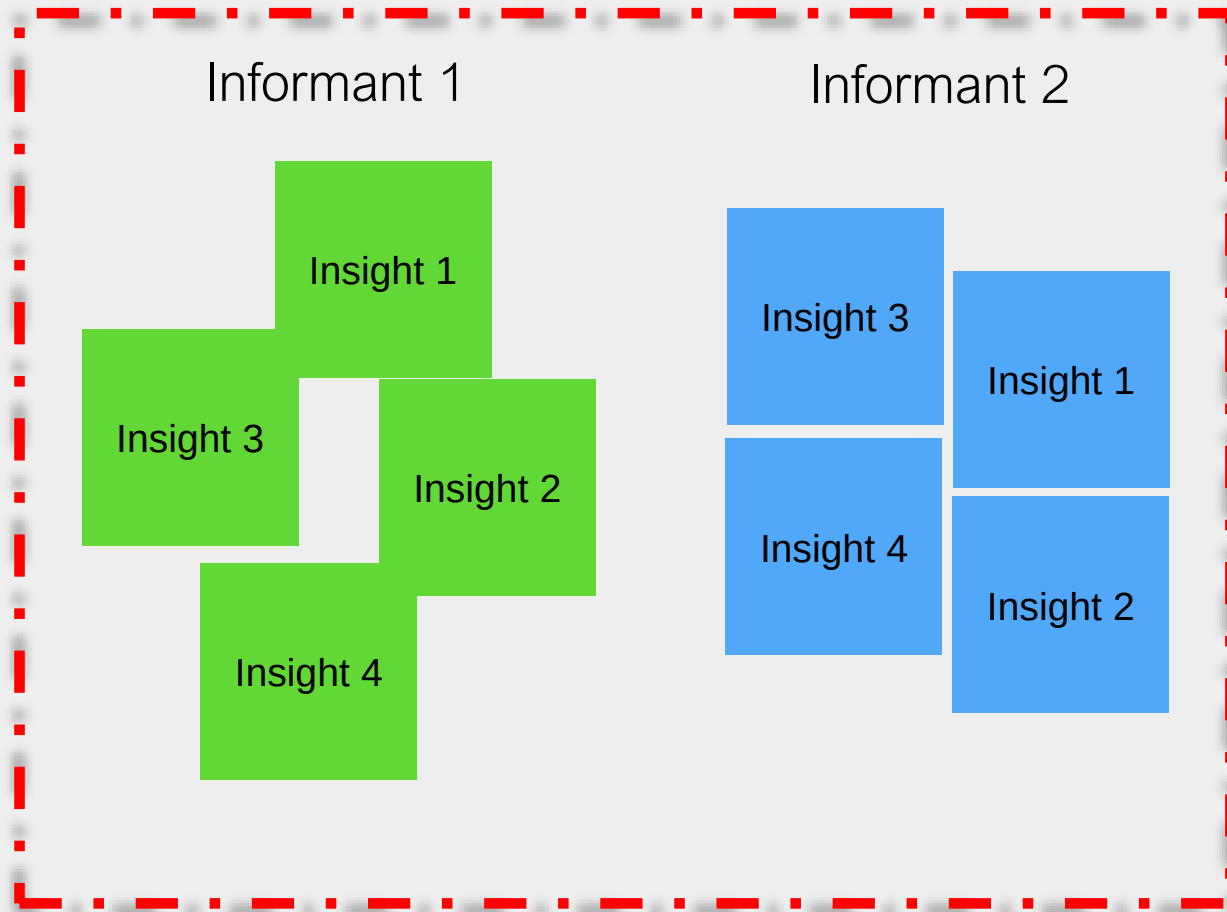
Goals

- Eat simply
- Not be overthinking about food

Frustration

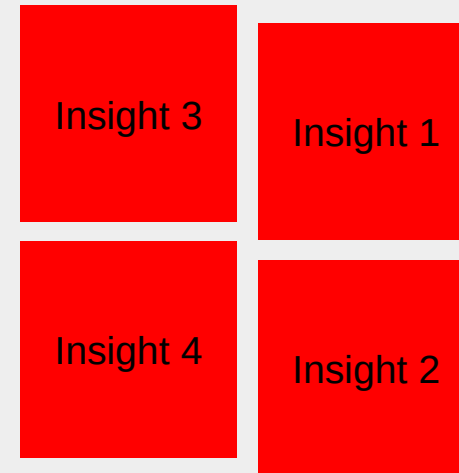
- Little time to eat with family
- Seldom be at home on weekdays

Persona 1



Persona 2

Informant 3



Persona 3

Informant 4



EMPATHY MAP

SEE? HEAR?

SAY & DO?

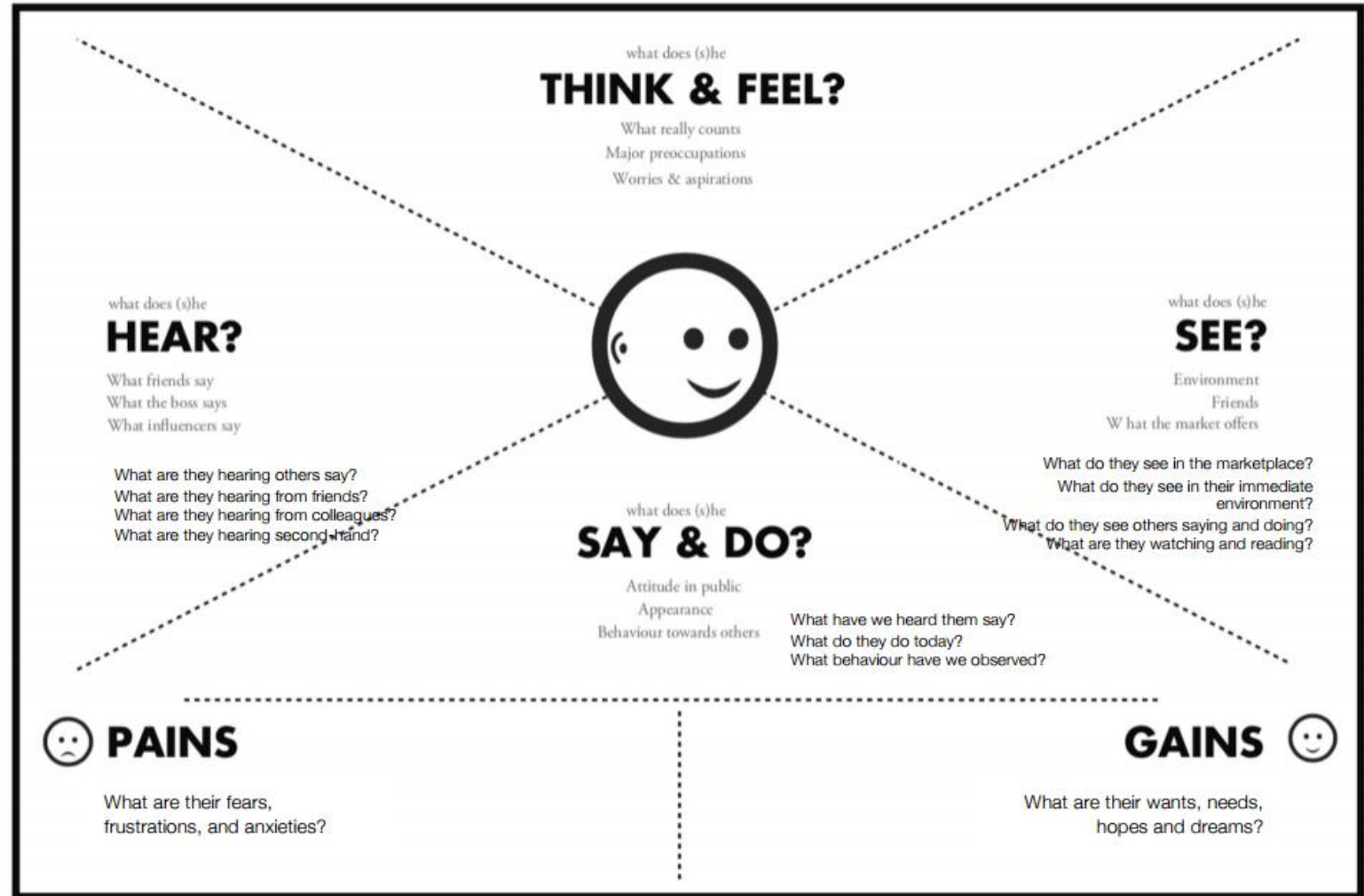
THINK & FEEL?

PAINS

- What are the pain points?
- Challenges?
- Problems or doubts
- Obstacles
- Frustrations

GAINS

- What do users hope to gain?
- What do users want to achieve?
- Goals, Needs
- What is success?
- How success is measured?



Empathy Map Canvas ©2017 David Grey

[Link to softcopy of templates for Empathy Map, JTBD and POEMS](#)



EMPATHY MAP

SEE? HEAR?

SAY & DO?

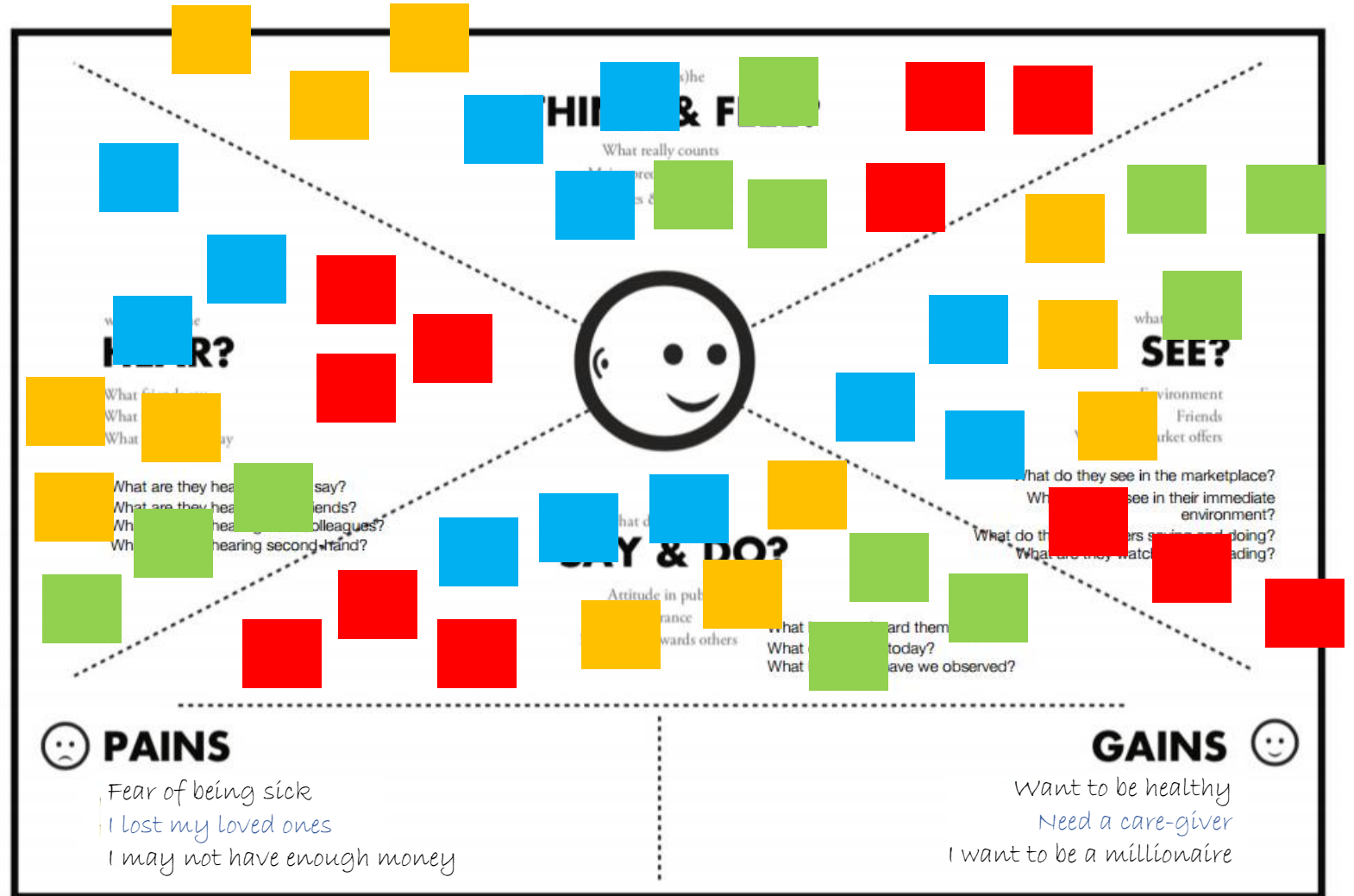
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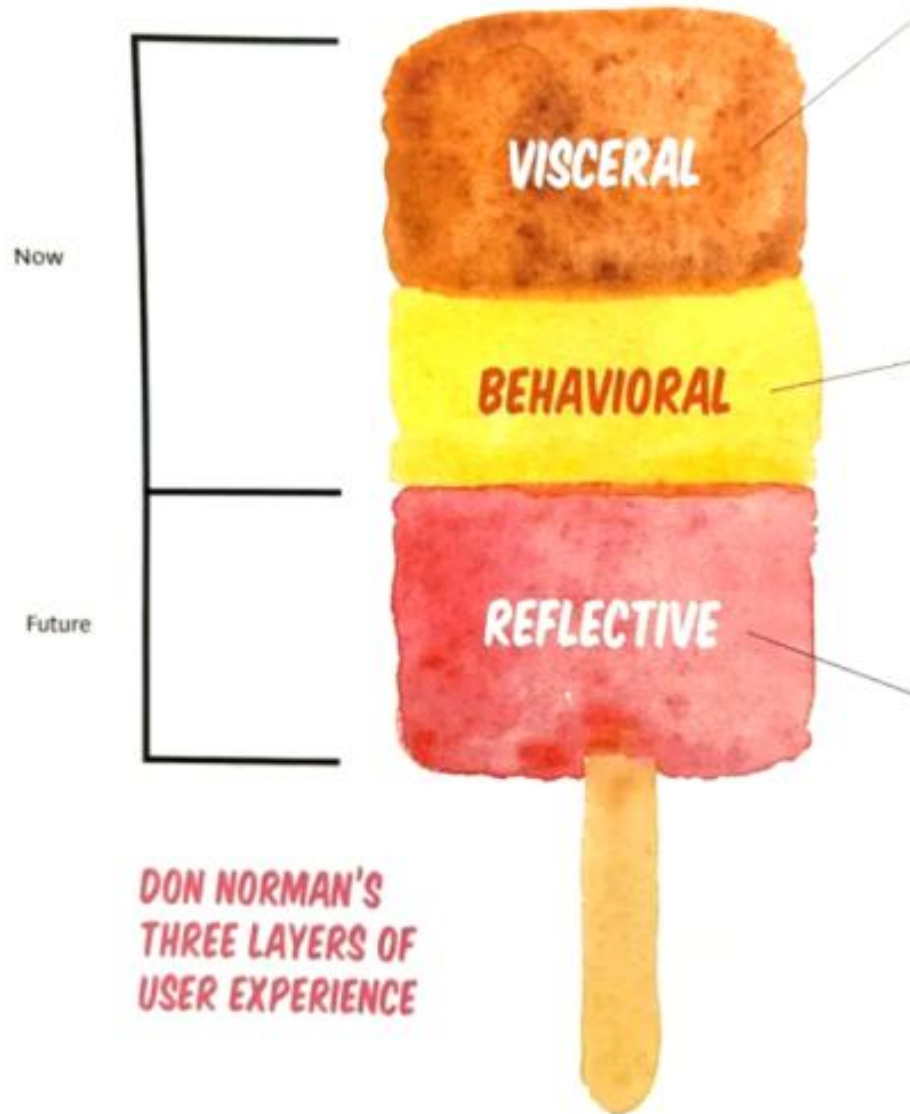
Experience Economy

The emotion wheel (Dr. Robert Plutchik's Wheel of Emotions) is a tool that helps to identify & label user's feelings by categorizing emotions into core, nuanced variations to facilitate better communication.



Experience Economy

This is how differently users respond to the user experience (or a story, a car ride, or anything else) with the passage of time (from the Beginning to the Future).



Beginning

At the visceral level, design stirs an immediate reaction to form, colour, texture and materials used.

Middle

At the behavioural level, design prompts a physical response or action.

End

At the reflective level, design engages memories and associations.

Experience Economy

*Adapted
Maslow's
Hierarchy
[of user
experience]*

Understanding user experience allows the designer to move up the hierarchy (i.e. from Functional to Pleasurable).

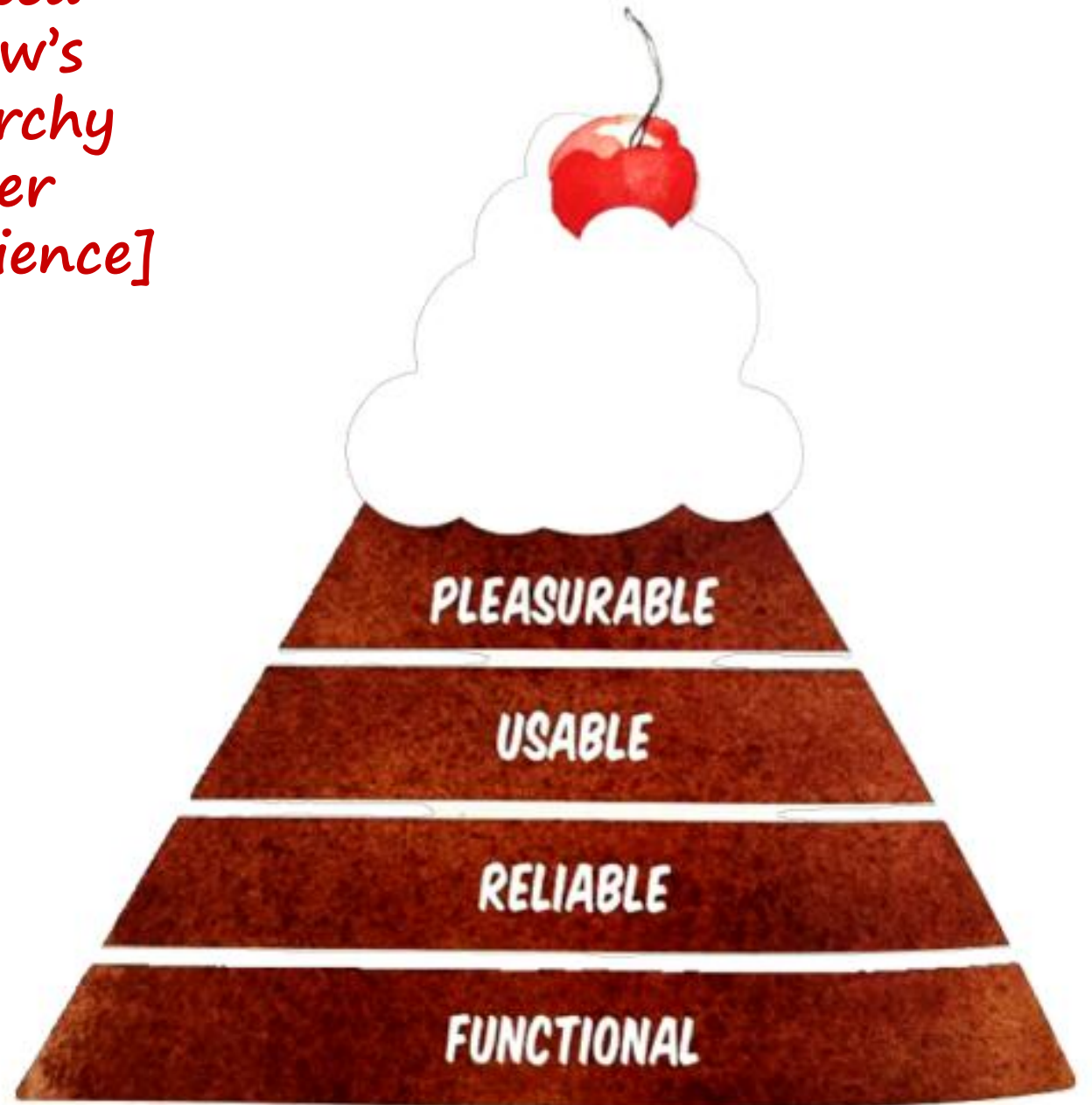
Functional: Does it work? If not, nothing else matters.

Reliable: Can users trust it to work consistently?

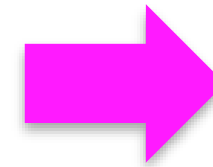
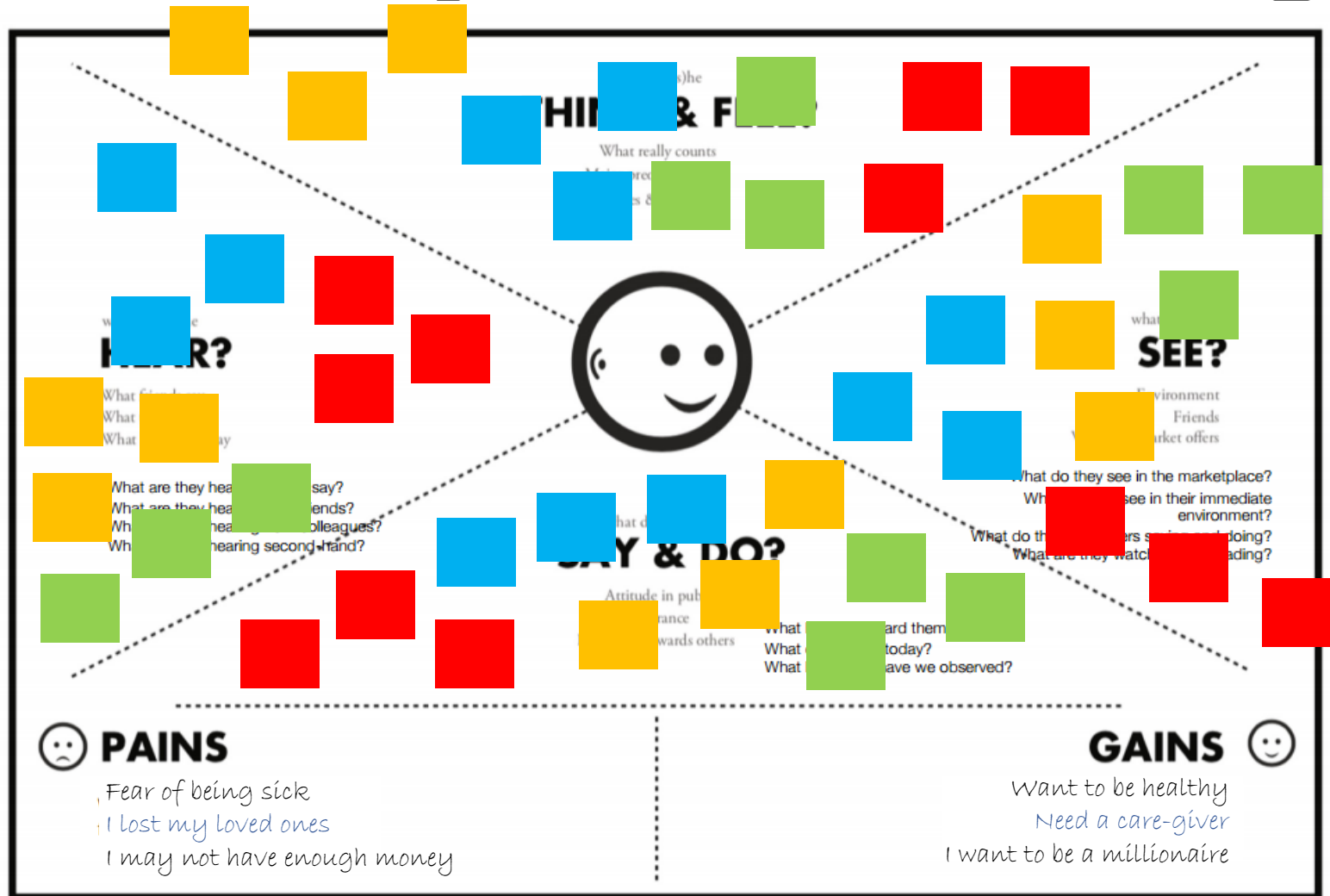
Usable: Is it easy and intuitive to use?

Pleasurable: Does it create moments of joy that users remember?

User delight is the cherry on top



Affinity Clustering



User Goals

Product Values

Product Features



User Goals – Why’s behind the JTBD

The desired outcomes or objectives that users aim to achieve by using a product or service.

Product Features – How the product delivers

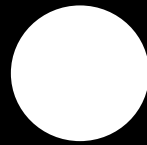
The specific functionalities or capabilities of a product.

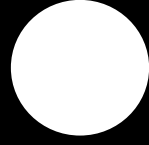
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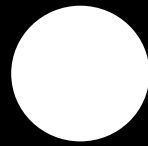


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Product Features – How the product delivers

The specific functionalities or capabilities of a product.



User Goals – Why's behind the JTBD

The desired outcomes or objectives that users aim to achieve by using a product or service.

User Goals (Why?)

A tech savvy user wants to communicate with friends and family easily.



Product Values – What the product delivers

The core benefits that a product brings to its users.

User Goals (Why?)

A tech savvy user wants to communicate with friends and family easily.

Product Values (What?)

A means for the user to communicate with friends and family using her mobile devices



Product Features – How the product delivers

The specific functionalities or capabilities of a product.

User Goal (Why?)

A tech savvy user wants to communicate with friends and family easily.

Product Value (What?)

A means for the user to communicate with friends and family using her mobile devices

Product Feature (How?)

A mobile app that allows the user to post pictures and also send/receive messages on-the-go



User Goals – Why's behind the JTBD

A student wants to learn a new hobby at her own free time.



User Goal (Why?)

A student wants to learn a new hobby at her own free time.

Product Value (What?)

A means for her to learn about the new hobby at her own pace.

Product Feature (How?)

An online learning platform that tracks her learning progress.



Innovation

A favourite word for both designers and engineers the development of new ideas or the improvement of existing ideas

but how?

How to be innovative?

Research? Ai?

Advancement in technology

Creativity

“Thinking” - How to think a new idea?

Activity Time!

~5 mins

A4 paper and writing materials!



Write down the key product you are going to innovate with E.g - Water bottle/Medicine dispenser/coaster

If your group does not have one, pick an item from below

Pen
Toothbrush
Keys

Exploring Design Oxymoron

Functional roots

A statement that describes the basic function of a product (It's core function)

Fan *An object that displaces air with electricity that powers a moving surface*

Fan *An object that displaces air*

Write their functional roots & Oxymoron statement
~10 mins

Draw and annotate!

- Contains a contradictory element
- Conveys a clear meaningful intent/objective
- Provides context how it's objective can be achieved.

Quantity vs Quality

Group Discussion
~10 mins

|

Each person, share your selected item, the design oxymoron statement and your concept

Bounce Phase
~8 mins



Pass your paper to another person, and now take 8mins to review the 8 ideas - Redraw with your own interpretation + suggestions

8mins total, 1mins each segment

**Reflection &
Group Discussion**
~10 mins

|

Everyone pass back the papers to the original owner and share your improvements with them

How is this useful to real projects?

Our case studies

